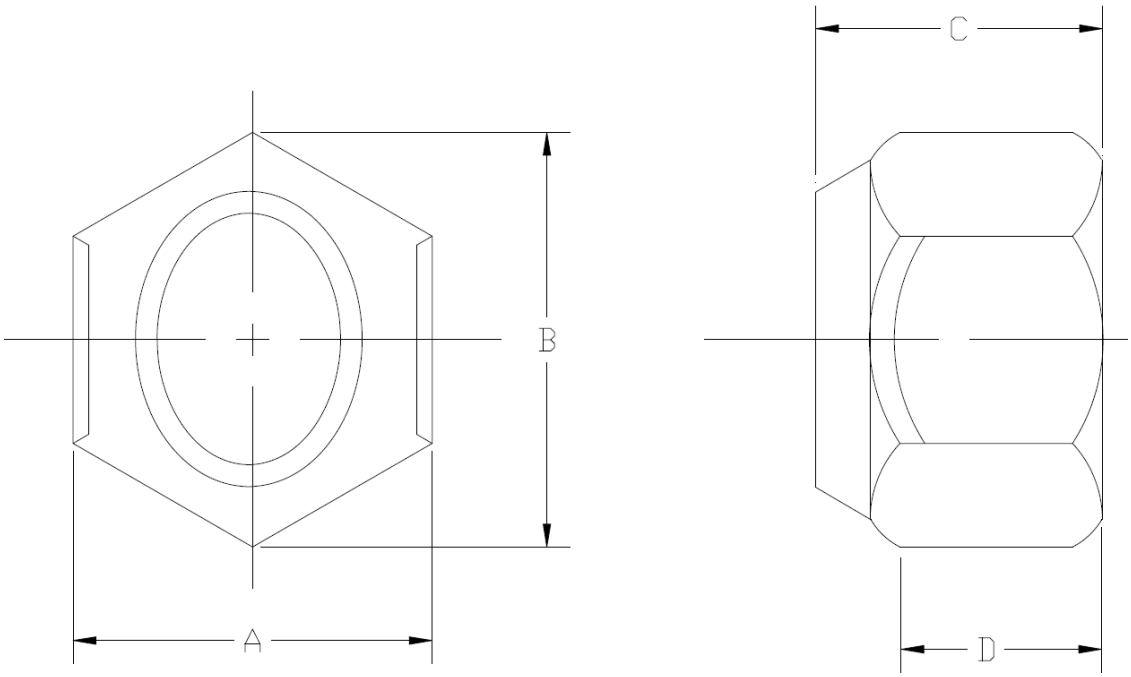
INCH-POUND
MIL-DTL-45913/4C
20 May 2021
SUPERSEDING
MIL-DTL-45913/4B
14 June 2011

DETAIL SPECIFICATION SHEET

NUT, SELF-LOCKING, HEXAGON, ALL-METAL LOCKING FEATURE, 250 DEGREES FAHRENHEIT, UNC/ UNF-3B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-45913.



(All dimensions are in inches.)

FIGURE 1. Nut, self-locking, hex, all-metal locking feature, UNC/UNF3B.

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REQUIREMENTS:

MATERIAL:

Steel and alloy steel shall be in accordance with SAE J995 grade 5 or 8, as specified in part-number.
Corrosion resisting steel (stainless steel) grades 303 and 316 or any alloy in accordance with ASTM F 594, Group 1 or Group 2, shall have a minimum ultimate tensile strength of 70 KSI and shall be non-magnetic.
Copper alloy (brass) shall be in accordance with SAE AMS4507, HR02 shall have a minimum ultimate tensile strength of 58 KSI and be of a half hard temper.
Aluminum alloy shall be 6061 in accordance with SAE AMS4117 and have a minimum ultimate tensile strength of 42 KSI or as specified in contract.

PLATING/FINISH:

Steel and alloy steel locknuts shall be:

Cadmium /1 plated per SAE-AMS-QQ-P-416 Type II, Class 2 or 3, or
Zinc-Nickel plated per MIL-PRF-32647, Type K16 or Type K20, or
Zinc plated per ASTM B633 /2 Type II, Fe/Zn 8, or
Black Oxide per MIL-DTL-13924, Class 1 or
Uncoated plain with a supplementary lubricant which is clean and dry to the touch.

Stainless steel locknuts shall be passivated SAE-AMS 2700 /3, Method 1, Type 6 or 7, Class 1, or black oxide finish per MIL-DTL-13924, Class 1.
Brass locknuts shall be black oxide finished per MIL-F-495, or nickel plated per SAE-AMS-QQ-N-290, Class 2, or uncoated plain with a brite finish.
Aluminum locknuts shall be anodized clear, no color per MIL-PRF-8625 (see Table I).

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred alternate.
/2 Type II passivation per ASTM B633 contains hexavalent chromium which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred alternates.
/3 Passivation per ASM2700 contains sodium dichromate which is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc with hex free passivation per ASTM F1941/F1941M Fe/Zn 12 or Zinc-Nickel plate per MIL-PRF-32647 Type K20 are the preferred alternates.

SURFACE TEXTURE. Shall be Ra 125 microinches in accordance with ASME B46.1.

THREADS. Shall be coarse or fine threaded, class 3B, as specified in part-number, in accordance with ASME B1.1.

WORKMANSHIP. Parts shall be manufactured and processed with a level of care and workmanship befitting this type of product intended for general purpose military application. Parts shall be free from dirt, grease, loose or deposited foreign materials, sharp edges, burrs, chips, physical malformations or anything that might prevent the part from reliably performing its intended function.

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PART NUMBER. The part number shall consist of the basic number of this specification sheet followed by a dash, a dash number from table I, a material code from table II, and a finish/plate code (when required) from table III.

EXAMPLE OF PART NUMBER:

M45913/4- 4C G5 QW

- QW — Finish Code (see Table III)
- G5 — Material Code (see Table II)
- 4C — Dash number from Table I to indicate size and thread.
- M45913/4- — Basic Specification Sheet Number followed by "-"

M45913/4-4CG5QW = Nut Self-Locking Hexagon, all-metal locking feature, .250-20 thread, Grade 5 steel, Zinc-Nickel, Type K20 plated.

TABLE I. Dash numbers and dimensions.

Dash No.	Thread Size	A		B	C		D
		Width Across Flats		Corners	Thickness		
		MAX	MIN	REF	MAX	MIN	REF
00F	.060-80 UNF-3B	.111	.104	.116	.055	.080	.045
01C	.073-64 UNC-3B	.251	.243	.268	.153	.133	.081
01F	.073-72 UNF-3B	.251	.243	.268	.153	.133	.081
02C	.086-56 UNC-3B	.251	.243	.268	.153	.133	.081
02F	.086-64 UNF-3B	.251	.243	.268	.153	.133	.081
03C	.099-48 UNC-3B	.251	.243	.268	.153	.133	.081
03F	.099-56 UNF-3B	.251	.243	.268	.153	.133	.081
04C	.112-40 UNC-3B	.251	.241	.275	.163	.087	.066
04F	.112-48 UNF-3B	.251	.241	.275	.163	.087	.066
05C	.125-40 UNC-3B	.251	.243	.268	.153	.133	.081
05F	.125-44 UNF-3B	.251	.243	.268	.153	.133	.081
06C	.138-32 UNC-3B	.313	.302	.344	.171	.102	.075
06F	.138-40 UNF-3B	.313	.302	.344	.171	.102	.075
08C	.164-32 UNC-3B	.345	.332	.378	.191	.117	.083
08F	.164-36 UNF-3B	.345	.332	.378	.191	.117	.083
010C	.190-24 UNC-3B	.376	.362	.413	.241	.117	.083
010F	.190-32 UNF-3B	.376	.362	.413	.241	.117	.083
012C	.216-24 UNC-3B	.439	.423	.482	.241	.148	.103
012F	.216-28 UNF-3B	.439	.423	.482	.241	.148	.103
4C	.250-20 UNC-3B	.439	.428	.488	.288	.212	.145
4F	.250-28 UNF-3B	.439	.428	.488	.288	.212	.145
5C	.312-18 UNC-3B	.502	.489	.557	.336	.258	.166
5F	.312-24 UNF-3B	.502	.489	.557	.336	.258	.166
6C	.375-16 UNC-3B	.564	.551	.628	.415	.320	.198
6F	.375-24 UNF-3B	.564	.551	.628	.415	.320	.198
7C	.437-14 UNC-3B	.688	.675	.768	.463	.365	.223
7F	.437-20 UNF-3B	.688	.675	.768	.463	.365	.223
8C	.500-13 UNC-3B	.752	.736	.840	.573	.427	.262
8F	.500-20 UNF-3B	.752	.736	.840	.573	.427	.262
9C	.562-12 UNC-3B	.877	.861	.982	.621	.473	.286
9F	.562-18 UNF-3B	.877	.861	.982	.621	.473	.286
10C	.625-11 UNC-3B	.940	.922	1.051	.731	.535	.329

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TABLE I. Dash numbers and dimensions – Continued.

10F	.625-18 UNF-3B	.940	.922	1.051	.731	.535	.329
12C	.750-10 UNC-3B	1.127	1.088	1.240	.827	.617	.382
12F	.750-16 UNF-3B	1.127	1.088	1.240	.827	.617	.382
14C	.875-9UNC-3B	1.314	1.269	1.447	.922	.724	.450
14F	.875-14 UNF-3B	1.314	1.269	1.447	.922	.724	.450
16C	1.000-8UNC-3B	1.502	1.450	1.653	1.018	.831	.513
16F	1.000-12 UNF-3B	1.502	1.450	1.653	1.018	.831	.513
18C	1.125-7UNC-3B	1.689	1.631	1.859	1.176	.939	.576
18F	1.125-12 UNF-3B	1.689	1.631	1.859	1.176	.939	.576
20C	1.250-7UNC-3B	1.877	1.812	2.066	1.272	1.030	.628
20F	1.250-12 UNF-3B	1.877	1.812	2.066	1.272	1.030	.628
22C	1.375-6UNC-3B	2.064	1.994	2.273	1.399	1.138	.681
22F	1.375-12 UNF-3B	2.064	1.994	2.273	1.399	1.138	.681
24C	1.500-6UNC-3B	2.252	2.175	2.480	1.526	1.245	.757
24F	1.500-12 UNF-3B	2.252	2.175	2.480	1.526	1.245	.757

TABLE II. Material coding.

MATERIAL CODE	
Steel Grade 5	G5
Steel Grade 8	G8
Stainless Steel 303	S3
Stainless Steel 316	S6
Brass	BB
Aluminum	AA

TABLE III. Finish coding. 1/

Cadmium	C
Zinc	Z
Black Oxide	B
Nickel	N
Zinc-Nickel (Type K16)	QX
Zinc-Nickel (Type K20)	QW
Anodized	no code (aluminum only)
Passivate	no code (stainless only)
Uncoated	no code

1/See requirements above for material and finish precautions.

NOTES:

1. Dimensions in inches, unless specified, tolerance: ± .016.
2. Interpret drawing in accordance with ASME Y14.5.
3. Break all Sharp edged .003 to .005 and remove all burrs and slivers.
4. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.
5. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.

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